11. Friday, July 11th. Hausdorff spaces. Countability axioms (§17, 21)

Definition: A topological space X is **Hausdorff** if for every $x \neq y \in X$, there are disjoint open sets $U, V \subseteq X$ with $x \in U$ and $y \in V$

Examples (1) If (X,d) is a metric space, then (X, T_d) is Hausdorff. Given $x \neq y \in X$, let $r = \frac{d(x,y)}{2} > 0$

Then $B(x,r) \cap B(y,r) = \emptyset$. In fact, if $z \in B(x,r) \cap B(y,r)$, then

$d(x,y) \leq d(x,z) + d(z,y) < r + r = d(x,y)$, ie $d(x,y) < d(x,y)$, a contradiction.

(2) Subspace of Hausdorff is Hausdorff.

(3) Let T_1 and T_2 be topologies on X with $T_1 \subseteq T_2$. If (X, T_1) is Hausdorff then (X, T_2) is Hausdorff

4) If $\{X_i \mid i \in \Lambda\}$ are all Hausdorff, then $\prod_{i \in \Lambda} X_i$ is Hausdorff both in box and product topologies. Since $T_{prod} \subseteq T_{box}$, enough to show it for T_{prod}. Given $x, y \in \prod X_i$, if $x \neq y$ then $x(j) \neq y(j)$ for some $j \in \Lambda$. Since X_j is Hausdorff then $\exists\ U_j, V_j$ open in X_j

with $x(j) \in U_j$, $y(j) \in V_j$ and $U_j \cap V_j = \emptyset$. Then $U_j \times \prod_{i \in \Lambda \setminus j\}} X_i$ and $V_j \times \prod_{i \in \Lambda \setminus j\}} X_i$ are disjoint open in τ_{Prod}, containing x and y respectively.

Proposition 1: Assume (x_n) is a sequence in a Hausdorff space X. If $x_n \to x$ and $x_n \to y$, Then $x = y$.

Proof: Assume for a contradiction that $x \neq y$. Let U, V disjoint open sets with $x \in U$ and $y \in V$. Since $x_n \to x$, $\exists\, N_1$ such that $x_n \in U \quad \forall n \geq N_1$

Since $x_n \to y$, $\exists\, N_2$ such that $x_n \in V \quad \forall n \geq N_2$

Let $N = \max\{N_1, N_2\}$ Then $x_N \in U \cap V$, contradicts $U \cap V = \emptyset$ □

Q: What about The converse?

Definition 2: A Topological Space is **first countable** if every point has a countable nbhd basis.

Example: The most important example To keep in mind is metric spaces!
Given $x \in (X, d)$ we consider the collection $\mathcal{B}_x = \{B(x, \frac{1}{n}) \mid n \in \mathbb{N}\}$. Since $\forall \varepsilon > 0 \;\exists\, n \in \mathbb{N}$ with $B(x, \frac{1}{n}) \subseteq B(x, \varepsilon)$ (Take any $n > \frac{1}{\varepsilon}$), This collection is a nbhd basis.

Lemma 3: If X is first countable, Then 1) Each $x \in X$ has a countable nbhd basis $\mathcal{B}_x = \{B_n \mid n \in \mathbb{N}\}$ such that $B_{n+1} \subseteq B_n \quad \forall n \in \mathbb{N}$.

2) If $(x_n)_{n \in \mathbb{N}}$ is any sequence such that $x_n \in B_n \ \forall n \in \mathbb{N}$, Then $x_n \to x$.

Proof: 1) Take a countable nbhd basis $\mathcal{B}'_x = \{U_n \mid n \in \mathbb{N}\}$

For each $n \in \mathbb{N}$, let $B_n = U_1 \cap U_2 \cap \cdots \cap U_n$. This is a nbhd of x And of course $B_{n+1} \subseteq B_n \ \forall n \in \mathbb{N}$. To prove These B_n form a basis at x, Let V be an open set containing x. Since $\mathcal{B}'_x$ is a nbhd basis, $\exists n \in \mathbb{N}$ such That $U_n \subseteq V$. Since $B_n \subseteq U_n$, we also have $B_n \subseteq V$, as desired.

2) HW! $\square$

Theorem 4: Let X be a first countable space. Then X is Hausdorff if and only if convergent sequences in X admit a unique limit.

Proof: ($\Rightarrow$) Is just Prop. 1.

($\Leftarrow$) Let $x \neq y$ in X. Since X is first countable, we can take nbhd basis

$\mathcal{B}_x = \{B_n \mid n \in \mathbb{N}\}$ and $\mathcal{B}_y = \{B_n' \mid n \in \mathbb{N}\}$ for x and y as in lemma 3.

Claim: $\exists\, n \in \mathbb{N}$ such that $B_n \cap B_n' = \emptyset$.

Otherwise, $\forall n \in \mathbb{N}$ $\exists\, z_n \in B_n \cap B_n'$. Consider the sequence $(z_n)_{n \in \mathbb{N}}$. Now apply lemma 3 (2)

Since $z_n \in B_n$ $\forall n$, we have $z_n \to x$

Since $z_n \in B_n'$ $\forall n$, we have $z_n \to y$

But limits in X are unique, hence $x = y$, a contradiction

□

Definition 5: Let X and Y be Top. spaces.

1) A subset $A \subseteq X$ is **sequentially closed** if for any sequence $(a_n)_{n \in \mathbb{N}}$ in A with $a_n \to x$ for some $x \in X$, we have $x \in A$.

2) A function $f: X \to Y$ is **sequentially continuous** at $x \in X$ if for every sequence (x_n) in X with $x_n \to x$ we have $f(x_n) \to f(x)$.